Funnel Analysis, A/B Testing Proposal & GA4/Hotjar Events Instrument

AKIB AHMED

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Executive Summary

- **The Opportunity:** Funnel analysis identifies the **Page View** → **Add-to-Cart** transition as the primary bottleneck, with a **49.05% drop-off rate**. This high-friction stage currently prevents nearly half of all high-intent traffic from progressing toward a purchase.
- **The Strategic Goal:** By implementing two targeted A/B tests focused on **Price Transparency** and **Decision Support**, we aim to reduce this drop-off to **43%**, achieving a **+12% relative uplift** in cart additions.
- **The Quantitative Impact:** Based on current traffic of **2,000 visitors**, this optimization is projected to capture **121 additional "Add-to-Cart"** events per cycle.
- **The Financial ROI**: Given the current **৳910.67** Average Order Value, successfully resolving this top-of-funnel leak represents a projected monthly revenue increase of **৳39,746**, with an annualized opportunity exceeding **৳476,000.**

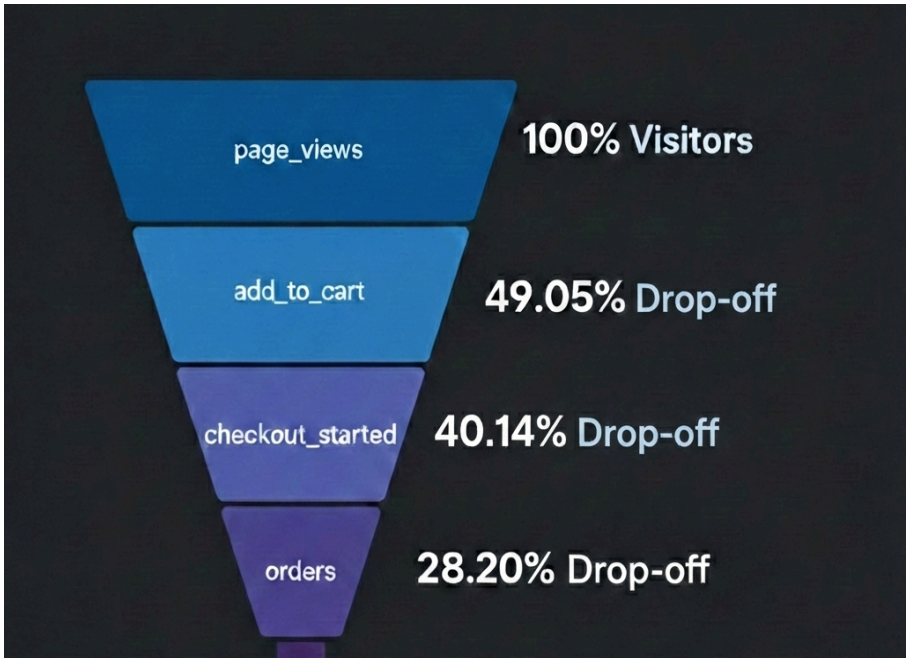
Problem Statement

1. **Data Insight:** Our database analysis identified two significant leaks: a **49.05% drop-off** at the `add_to_cart` stage and a **40.14% drop-off** at the `checkout_started` stage.
2. **The "Leak":** We are losing nearly half of our high-intent users immediately after they add a book to their cart, indicating a major friction point in the transition to the checkout flow.

Goals

3. **Primary Metric:** Product View-to-Add to cart Conversion Rate.
4. **Secondary Metric:** Page Scroll Depth (to measure engagement with the new widgets).

Funnel Analysis



Key Findings

- Add-to-Cart Drop-off: **49.05%** (Highest leakage)
- Checkout Start Drop-off: **40.14%**
- Purchase Drop-off: **28.20%**

Improving early-stage conversion will compound downstream gains.

Hypothesis & Proposed A/B Tests

Test 1: Price & Delivery Transparency

- **Hypothesis:** We believe that users are dropping off because they fear hidden shipping costs. By moving the "**Estimated Delivery Date**" and "**Free Shipping Eligibility**" directly next to the price on the product page, we will reduce abandonment.
- **Control (A):** Shipping details only visible in the cart or footer.
- **Variant (B):** "Ships tomorrow - Free shipping on orders over “BDT" placed immediately below the "Add to Cart" button.

Expected Uplift:

- **Current Baseline:** add_to_cart conversion rate (after the initial page_view) is currently **50.95%** (derived from the **49.05% drop-off**).

- **The Goal:** We want to reduce that drop-off. If we successfully lower the drop-off from **49.05%** to **43%**, our new add_to_cart rate becomes **57%**.

- **The Relative Uplift:** (57% - 50.95%) / 50.95% approx = **+12%** Relative Uplift

Test 2: Introduce Compatibility/Comparison

- **Hypothesis:** We believe that users are leaving to compare different editions (Hardcover vs. Kindle vs. Paperback) or to check prerequisites. If we add a "**Which Edition is right**

for you?" comparison table directly on the product page, we reduce the need for them to leave the page to research.

- **Control:** Standard product description and edition selector.
- **Variant:** A clear comparison table highlighting the benefits of each format (e.g., "Best for students," "Instant access," "Collector's edition").
- **Expected Uplift: +10%** relative increase in `add_to_cart`.

Business Impact

Metric	Baseline	Target (Test 1)	Target (Test 2)
Add to Cart Rate	50.95%	57.00%	56.05%
Absolute Gain	—	+6.05%	+5.10%
Relative Uplift	—	**+11.87%**	**+10.01%**

Instrument GA4/Hotjar events

GA4 Strategy :

1. Validation for Test #1: Price & Delivery Transparency

We need to prove that seeing the shipping cost earlier is what actually drives the conversion.

- **Event:** pdp_transparency_view
- Trigger: An Element Visibility trigger (via GTM) that fires when the "Ships Tomorrow / Free Shipping" microcopy is in the viewport for >2 seconds.
- **Event:** shipping_threshold_delta
- Parameter: distance_to_free_shipping (Numeric).
- Logic: Tracks the value in the dynamic progress bar.
- Success Metric: Correlation Analysis. We will run a segment in GA4 to see if the Checkout-to-Order rate (currently 28.20% drop-off) improves for users who interacted with transparency elements, proving we are qualifying traffic earlier.

2. Validation for Test #2: Comparison Matrix

- **Event:** matrix_research_engagement
- Parameter: interaction_type (e.g., "row_hover", "column_sort", "feature_click").
- **Event:** edition_selection_intent
- Logic: Fires when a user clicks a specific edition inside the matrix versus the standard drop-down.
- Success Metric: Session Quality. We expect users who interact with the matrix to have a higher session_start to add_to_cart ratio, validating that the tool solved "Decision Paralysis".

Hotjar Strategy:

1. Strategic Heatmap Comparison (Split Testing)

- Metric: Engagement Zone Shift. In Variant B (Matrix), we should see a "Hot Zone" over the comparison table. If the heat remains only on the "Add to Cart" button but the conversion hasn't moved, the table is being ignored.
- Move Map: Track how much time users spend hovering over the price vs. the shipping transparency text to see which "Incentive Trigger" is actually processed.

2. Friction-Specific Session Recordings

We will set up Recording Filters to watch the "Failures" specifically:

- Filter 1: Users who spent >60 seconds on the page + had "Rage Clicks" + Exit. This tells us where the logic of our new content (Matrix or Shipping) is breaking.

- Filter 2: Users who engaged with the "Urgency Trigger" but still dropped off at the 40.14% Checkout Started stage. This helps us see if the urgency feels "fake" or if it creates a panic-exit rather than a purchase.

3. Qualitative "Micro-Surveys"

- Trigger: 50% Scroll Depth OR 45 seconds of inactivity.
- Question: "Is there a specific detail about the [Book Edition] you couldn't find here?"
- Goal: This directly validates Test #2. If users still say "I don't know the difference between editions," then our Matrix design needs iteration.

Summary Table

Hypothesis	Primary Tracking Tool	Metric of Truth

Transparency (Test 1)	**GA4 Custom Event**	add_to_cart **Conversion Rate (Target: 57%)**
Comparison (Test 2)	**Hotjar Heatmaps**	**Click-through rate on the Matrix vs. Page Exit**
Trust/Urgency	**GA4 + GTM**	checkout_started **consistency (Ensuring we don't increase the 40.14% leak)**
Overall ROI	**GA4 E-commerce**	**Revenue Lift vs. Baseline (Target: +$NULL/mo)**

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